

Binary Response Models and Sample Selection

An Applied Econometrics Portfolio Project in Stata

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Applied-Econometrics-with-Stata

Methods: linear probability models, heteroskedasticity-robust inference, probit estimation, marginal effects, classification analysis, nonlinear probability models, and Heckman sample-selection correction.

Executive Summary

This project studies two common empirical challenges in applied microeconometrics: modeling binary outcomes and correcting for nonrandom sample selection. The first application analyzes arrest outcomes using a linear probability model and probit specifications. The second application estimates a women's wage equation when wages are observed only for labor-force participants.

In the arrest application, a higher probability of conviction is associated with a lower probability of arrest. The linear probability model implies that increasing the conviction probability from 0.25 to 0.75 reduces arrest probability by approximately 7.7 percentage points, while the probit model implies a reduction of about 10.2 percentage points. A nonlinear probit model provides evidence that the effect varies with the conviction probability.

In the wage application, the Heckman selection correction produces a statistically significant inverse Mills ratio. The corrected return to experience is smaller than the selected-sample OLS estimate, indicating that ignoring labor-force participation leads to upward-biased wage-equation estimates.

Finding	Evidence	Interpretation
Conviction risk predicts fewer arrests	pcnv coefficient in robust LPM = -0.154	A 0.50 increase in pcnv lowers arrest probability by about 7.7 percentage points
Probit effect is larger	Predicted probability: 0.398 at pcnv=0.25 and 0.296 at pcnv=0.75	Estimated discrete change is -10.2 percentage points
Baseline classification is imbalanced	Overall accuracy 72.7%; sensitivity for arrests 10.3%	Accuracy is driven by correctly predicting the majority non-arrest class
Nonlinearity matters	Quadratic terms jointly significant	The effect of conviction probability changes across its range
Selection bias affects wage estimates	Inverse Mills ratio = -0.409, $p=0.001$	OLS on working women overstates the return to experience

1. Economic Motivation and Binary-Choice Framework

A simple household-choice model illustrates how binary participation decisions can arise from optimization. A household chooses consumption and charitable giving subject to a budget constraint. The optimal quantity of giving is zero when the marginal utility parameter does not exceed the effective price of giving, and positive otherwise.

When the unobserved marginal utility component is modeled as an exponential index, the probability of giving becomes a standard latent-index binary-choice model. Depending on the assumed distribution of the normalized disturbance, the empirical model can be estimated using probit or logit methods.

2. Arrest Outcomes: Linear Probability Model

The empirical arrest model uses 2,725 observations. The dependent variable equals one when an individual was arrested in 1986. Explanatory variables include the probability of conviction, average sentence length, total prior prison time, prison time in 1986, income, and demographic controls.

Variable	Robust LPM coefficient	Interpretation
Probability of conviction (pcnv)	-0.1544	Higher conviction probability is associated with a lower arrest probability
Average sentence length	0.0035	Not statistically significant
Total prior prison time	-0.0021	Not statistically significant
Prison time in 1986	-0.0216	Longer incarceration is associated with lower arrest probability
Income in 1986	-0.00122	Higher income is associated with lower arrest probability
Black	0.1617	Positive conditional difference in arrest probability
Hispanic	0.0893	Positive conditional difference in arrest probability

The model R-squared is 0.0824. Because the linear probability model is heteroskedastic by construction, heteroskedasticity-robust standard errors are used for inference.

2.1 Joint Significance

Average sentence length and total prior prison time are not jointly significant. The robust joint test yields $F(2, 2716)=0.18$ with $p=0.832$. The substantive conclusion is unchanged under conventional standard errors.

2.2 Discrete Effect of Conviction Probability

The robust LPM coefficient on pcnv is -0.1544. Increasing pcnv from 0.25 to 0.75 therefore changes the fitted probability by $0.50 \times -0.1544 = -0.0772$. This corresponds to an estimated reduction of 7.7 percentage points.

3. Probit Estimation and Prediction

A probit model imposes fitted probabilities between zero and one and allows covariate effects to depend on the location of an observation in the latent-index distribution.

Scenario	Predicted arrest probability
pcnv = 0.25	0.3979
pcnv = 0.75	0.2962
Discrete change	-0.1017

Holding the remaining covariates at the values used in the original analysis, increasing pcnv from 0.25 to 0.75 reduces predicted arrest probability by about 10.2 percentage points. The qualitative conclusion agrees with the linear probability model, but the probit effect is larger in magnitude.

3.1 Classification Performance

Actual outcome	Predicted 0	Predicted 1	Class accuracy
No arrest	1,903	67	96.6%
Arrest	677	78	10.3%
Total	2,580	145	72.7% overall

The 0.50 classification threshold produces high overall accuracy, but this metric is misleading because arrests are less common. The model identifies only 10.3% of actual arrests. Precision-recall tradeoffs, alternative thresholds, and class-sensitive metrics would provide a more informative assessment.

4. Nonlinear Probit Specification

The probit model is extended with squared terms for conviction probability, prison time, and income. The squared terms are individually significant and jointly significant, indicating that a purely linear index is too restrictive.

Nonlinear term	p-value	Result
pcnv squared	0.002	Significant
ptime86 squared	<0.001	Significant
inc86 squared	0.041	Significant

The estimated latent-index relationship in pcnv is concave. Using the reported coefficients, the turning point is approximately 0.126. Above this value, the marginal effect of conviction probability on arrest probability is negative.

5. Sample Selection: Theory

The Heckman framework addresses cases in which the outcome is observed only for a nonrandomly selected subsample. In the wage application, wages are observed only for women who participate in the labor force.

Under the standard normal selection model, the conditional outcome equation includes the inverse Mills ratio. The original assignment also considers a generalized conditional-mean structure with both a linear and quadratic function of the selection disturbance. In that extension, two correction terms are required: the inverse Mills ratio and its interaction with the selection index.

Evidence of selection bias is obtained by testing whether the correction terms are jointly zero. In the standard Heckman model, this reduces to testing the coefficient on the inverse Mills ratio.

6. Women's Wages: Heckman Selection Application

The MROZ application contains 753 women, of whom 428 have observed wages and 325 are not employed. The outcome equation models log wages as a quadratic function of labor-market

experience. The selection equation includes education, age, young and older children, non-wife income, and parental and spouse education.

6.1 Labor-Force Participation Equation

Education significantly increases labor-force participation. Age, the presence of children younger than six, and non-wife income significantly reduce participation. These patterns are consistent with standard labor-supply theory.

6.2 Wage Equation and Selection Correction

Variable	Selected-sample OLS	Heckman two-step
Experience	0.0476	0.0424
Experience squared	-0.00102	-0.00085
Inverse Mills ratio	-	-0.409

Experience has a positive effect on wages, while the negative quadratic term implies diminishing returns. The inverse Mills ratio is negative and statistically significant ($p=0.001$), providing evidence of nonrandom selection.

The selected-sample OLS estimate of the return to experience exceeds the Heckman estimate. This difference indicates that estimating wages only among working women overstates the return to experience when selection is ignored.

7. Methodological Lessons and Limitations

- Linear probability models are easy to interpret but require robust inference and may generate fitted probabilities outside the unit interval.
- Overall classification accuracy can be uninformative when the outcome is imbalanced. Sensitivity, specificity, precision, recall, and threshold analysis should also be reported.
- Probit coefficients are latent-index parameters. Discrete changes and marginal effects are more interpretable than raw coefficients.
- Quadratic terms should be interpreted jointly with their associated linear terms; marginal effects can vary across the support of a regressor.
- The Heckman model relies on distributional assumptions and benefits from credible exclusion restrictions in the selection equation.
- A significant inverse Mills ratio indicates selection, but the economic sign of selection should be interpreted cautiously and in conjunction with the estimated correlation parameter.

8. Reproducibility

The accompanying Stata do-file implements the empirical workflow using relative project paths, input checks, model storage, robust inference, marginal effects, prediction, nonlinear specifications, and both manual and built-in Heckman estimators.

Recommended repository structure:

03_Binary_Response_and_Sample_Selection/

README.md

03_binary_response_and_sample_selection.do

Binary_Response_and_Sample_Selection_Report.pdf

data/README.md

Conclusion

The project demonstrates how nonlinear probability models and sample-selection corrections improve empirical analysis when ordinary least squares is poorly suited to the data-generating process. Both the LPM and probit models indicate that a higher probability of conviction is associated with a lower probability of arrest, while the nonlinear specification shows that this relationship varies across the conviction-risk distribution. In the wage application, the Heckman correction provides strong evidence that labor-force participation is nonrandom and that selected-sample OLS overstates the return to experience.